Common Quarterly Examination - September 2026

18/09/26

Standard 12

Time Allowed: 3.00 Hours **PHYSICS** Maximum Marks: 70

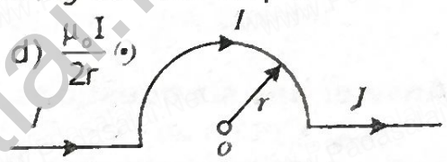
Part-I

Choose the best option: **15×1=15**

1) If Voltage applied on a capacitor is increased from V to 2V, choose the correct conclusion
   a) Q remains the same, C is doubled b) Q is doubled, C doubled
   c) C remains same, Q doubled d) Both Q and C remains same
2) Which one of the following is non-polar molecule?
   a) N_2O b) CO_2 c) HCl d) NH_3
3) A wire of resistance 2 ohms per meter is bent to form a circle of radius 1m. The equivalent resistance between its two diametrically opposite points A and B as shown in the figure is
   a) $\pi\Omega$ b) $\frac{\pi}{2}\Omega$ c) $2\pi\Omega$ d) $\frac{\pi}{4}\Omega$



4) _____ is used for electric fuses when current rating is above 15 A.
   a) An alloy of lead tin b) An alloy of Nickel and Chromium
   c) Copper fuse d) Manganin fuse
5) The magnetic field at the centre O of the following current loop is
   a) $\frac{\mu_0 I}{4r}\otimes$ b) $\frac{\mu_0 I}{4r}\odot$ c) $\frac{\mu_0 I}{2r}\otimes$ d) $\frac{\mu_0 I}{2r}\odot$



6) A non-conducting charged ring carrying a charge of q, mass m and radius r is rotated about its axis with constant angular speed ω. Find the ratio of its magnetic moment with angular momentum is _____.
   a) $\frac{q}{m}$ b) $\frac{2q}{m}$ c) $\frac{q}{2m}$ d) $\frac{q}{4m}$
7) A wire of length l carrying a current I along the y direction is kept in a magnetic field given by $\vec{B} = \frac{B}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$. The magnitude of Lorentz force acting on the wire is _____.
   a) $\sqrt{\frac{2}{3}}\beta Il$ b) $\sqrt{\frac{1}{3}}\beta Il$ c) $\sqrt{2}\beta Il$ d) $\sqrt{\frac{1}{2}}\beta Il$
8) In a series RL circuit, the resistance and inductive reactance are the same. Then the phase difference between the voltage and current in the circuit is
   a) $\frac{\pi}{4}$ b) $\frac{\pi}{2}$ c) $\frac{\pi}{6}$ d) zero
9) In a transformer, the number of turns in the primary and secondary are 410 and 1230 respectively. If the current in primary is 6A, then that in the secondary coil is _____.
   a) 2 A b) 18 A c) 12 A d) 1 A
10) To reduce the eddy current loss, the core of the transformer is made up of _____.
   a) Wires of larger diameter b) Insulated thin laminas
   c) Group of insulated wires d) Insulated toroid
11) Fraunhofer lines are an example of _____ spectrum.
   a) Line emission b) Line absorption
   c) Band emission d) Band absorption
12) In an electromagnetic wave travelling in a free space, the rms value of electric field is 3 Vm^{-1}. The peak value of magnetic field is _____.
   a) 1.414×10^{-8} T b) 1.0×10^{-8} T c) 2.828×10^{-8} T d) 2.0×10^{-8} T

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13) Which of the following is correct reason for star twinkling?
a) Reflection b) Total internal reflection
c) Refraction d) Polarization

14) The radius of curvature of a curved surface of a thin plano convex lens is 10 cm and the refractive index is 1.5. If the plane surface is silvered, then the focal length will be _____.
a) 5 cm b) 10 cm c) 15 cm d) 20 cm

15) The focal length of a convex lens _____ when red light is replaced by violet light.
a) becomes zero b) decreases
c) increases d) remains constant

Part-II

Answer any six questions. Question No.24 is compulsory: 6×2=12

16) What is meant by quantisation of charges?
17) Define current density.
18) State Ampere's circuital law.
19) Why is the path of a charged particle not a circle when its velocity is not perpendicular to the magnetic field?
20) Mention the ways of producing induced emf.
21) Distinguish between step up and step down transformers?
22) What is displacement current?
23) Why sky appears blue?
24) A coil of 200 turns carries a current of 0.4A. If the magnetic flux of 4m Wb is linked with each turn of the coil, find inductance of the coil.

Part-III

Answer any six questions. Question No.33 is compulsory: 6×3=18

25) Obtain an expression for the torque experienced by an electric dipole due to a uniform electric field.
26) Explain the equivalent resistance of parallel resistor network.
27) Compare dia, para and ferro magnetism.
28) Discuss the conversion of galvanometer into an ammeter.
29) An inductor of inductance L carries an electron current i. How much energy is stored while establishing the current in it?
30) Write down any six properties of electromagnetic waves.
31) A magnetron in a microwave oven emits electromagnetic waves with frequency f = 2450 MHz. What magnetic field strength is required for electrons to move in circular path with this frequency?
32) Obtain the equation for apparent depth.
33) A rectangular coil of area 70 cm^2 having 600 turns rotates about an axis perpendicular to a magnetic field of 0.4 Wbm^{-2}. If the coil completes 500 revolutions in a minute, calculate the instantaneous emf when the plane of the coil is (i) Perpendicular to the field (ii) Parallel to the field and (iii) Inclined at 60° with the field.

Part-IV

Answer all questions: 5×5=25

34) Calculate the electric field due to an electric dipole on its axial line. **(OR)** How the emf of two cells are compared using potentiometer?
35) Deduce the relation for the magnetic field at a point due to an infinitely long straight conductor carrying current using Biot-Savart law. **(OR)** Discuss the principle, construction and working of cyclotron with diagram.
36) Explain the principle, construction and working of transformer. **(OR)** Derive an expression for phase angle between applied voltage and current in a series RLC circuit.
37) Write down Maxwell's equations in integral form. **(OR)** Explain the types of emission spectrum.
38) Describe the Fizeau's method to determine the speed of light. **(OR)** Derive the equation for angle of deviation produced by prism and thus obtain the equation for refractive index of material of the prism.

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